The Chinese Remainder Theorem (due to Sunzi)

- **Thm** (The "Chinese Remainder Thm.") Suppose $m_1, m_2, \dots, m_k$ are pairwise rel. prime and that $c_1, c_2, \dots, c_k$ are integers.
  - Then there exists a unique (up to multiplying by n = $m_1 \cdot m_2 \cdot \dots \cdot m_k$) solution to system of equations:
    - $\{x \cong c_i \; mod \; m_i\}$
- **Proof** (existence):
  - Again, let $n = m_1 \cdot m_2 \cdot \dots \cdot m_k$
  - For each $i \in \{1, \dots, k\}$ define $n_i = n/m_i$
  - Notice that n_i is rel. prime to m_i (why?)
    - By easy inductive argument – notice that if a, b and c are rel. prime then so are a and $c \cdot b$
  - So let $\{a_i\}$ be set of mult. inverses wrt $\{n_i\}$ – i.e., let $a_i \cong \; n_i^{-1} mod \; m_i$ (why do these exist?)
  - Recall what this means: it's the unique a_i so that $a_i n_i \cong 1 \; mod \; m_i$
  - Define $x_i = a_i n_i$ for all i
  - Then claim: $x = c_1 x_1 + c_2 x_2 + \cdots + c_k x_k$ is the solution to the system!
  - To see this, look any i-th equation in the system: $\mathrm{x} \cong c_i \; mod \; m_i$, then:
    - Take i-th term of x: $c_i x_i = c_i \cdot a_i n_i$ (by definition of x_i)
      - **Claim:** $c_i \cdot a_i n_i \cong c_i \; mod \; m_i$ -- why?
      - Since $a_i \cong n_i^{-1} \; mod \; m_i$ so $c_i \cdot a_i n_i \cong c_i \cdot n_i \cdot n_i^{-1} mod \; m_i = c_i \cdot 1 \; mod \; m_i = c_i \; mod \; m_i$
    - But if we consider j-th term, for $j \neq i$, it's $c_j x_j$
      - Claim: $c_j x_j \cong 0 \; mod \; m_i$
      - Since $c_j x_j = c_j a_j n_j$ (by how we defined x_j)
        - $\cong 0 \; mod \; m_i$ (why??)

          Because by definition of n_j, $m_i | n_j \Rightarrow m_i | c_j a_j n_j$
  - QED

The Chinese Remainder Thm. (uniqueness)

- **Recall: Thm** (The "Chinese Remainder Thm.") Suppose $m_1, m_2, \ldots, m_k$ are pairwise rel. prime and that $c_1, c_2, \ldots, c_k$ are integers. Also, let $n = m_1 \cdot m_2 \cdot \ldots \cdot m_k$
  - Then there exists **a unique** (up to multiplying by n) solution to system of equations:
    - $\{x = c_i \; mod \; m_i\}$
- **Pf** (uniqueness)
  - Suppose not, so I have two solutions x_1 and x_2 so that for all i, $\{x_1 = c_i \; mod \; m_i\}$ and $\{x_2 = c_i \; mod \; m_i\}$.
  - Then $x_1 - x_2 = 0 \; mod \; m_i$ for all i
  - So for all i, $m_i | (x_1 - x_2)$ by definition
  - By assumption $m_i' s$ are pairwise relatively prime, so product, $n = m_1 m_2 \ldots m_k$ must also divide $(x_1 - x_2)$
    - by thm. (*) of prior slide
  - Thus $x_1 = x_2 \; mod \; n$
    - By definition of mod!
- To test understanding, work out examples of solving equations via the CRT! i.e., Ex. 4.21 of Kurtz